

Day 46 - QoS (pt 1)

Wednesday, September 9, 2026 8:37 PM

QoS Info

IP Phones

Traditional phones operate over the Public Switched Telephone Network (PSTN) / Plain Old Telephone Service (POTS).

- **IP Phones** are phones that use **VoIP (Voice over IP)** technologies to make phone calls over an IP network, such as the internet.
- **IP Phones** are connected to a switch just like any other end host.
- **IP Phones** have an internal 3-port switch, and they are:
 - An '**uplink**' port to the external switch.
 - A '**downlink**' port to the PC.
 - A port that connects internally to the phone itself.
- This allows the PC and the **IP Phone** to share a single switch port. Traffic from the PC passes through the **IP Phone** to the switch.
- It is recommended to separate '**voice**' traffic (coming from the **IP Phone**) and '**data**' traffic (coming from the PC) by placing them in separate VLANs.
 - This can be done by setting up a **Voice VLAN** in Switchport **access mode**.
 - **Voice** traffic will be sent tagged in the specified VLAN while **data** traffic will be sent untagged, as is the usual for an **access** port.
 - Although the interface sends/receives traffic from two separate VLANs, it is not considered a **trunk** port. It is considered an **access** port.

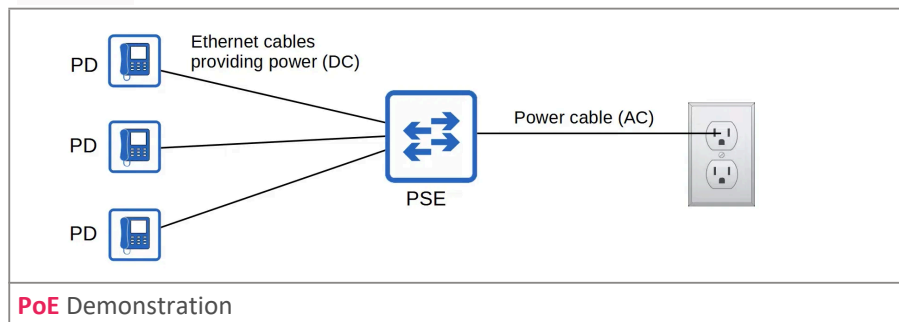
IP Phones Configuration

- `switchport voice vlan VLAN-ID`
 - The end host will send **voice** traffic tagged in the specified VLAN.
 - This command needs to be configured while the switchport is in **access mode** (through `switchport mode access`)

Power over Ethernet (PoE)

Power over Ethernet (PoE) is a feature that allows **Power Sourcing Equipment (PSE)** to provide power to **Powered Devices (PD)** over an Ethernet cable.

- The **PSE** is usually a switch and the **PDs** are devices like **IP Phones**, **IP Cameras**, **Wireless Access Points**, etc.
- The **PSE** receives **AC** power from an outlet and converts it to **DC** power, it then supplies that converted **DC** power to the **PDs**



- **PoE** has a process to determine if a connected device needs power, and how much power it needs. It does this as too much electrical current can damage electrical devices.
 - When a device is connected to a **PoE**-enabled port, the **PSE** (switch) sends low power signals, monitors the response, and determines how much power the **PD** needs.
 - If the device needs power, the **PSE** supplies the power to allow the **PD** to boot.

- The **PSE** continues to monitor the **PD** and supply the required amount of power.

PoE - Power Policing & Configuration

Power policing can be configured to prevent a **PD** from taking too much power.

- **power inline police OR power inline police action err-disable**
 - Configures **power policing** with the default settings.
 - It will disable the port and send a **Syslog** message if a **PD** draws too much power.
 - The interface will be put in an '**error-disabled**' state and can be re-enabled by running **shutdown** then **no shutdown**
- **power inline police action log**
 - Configures **power policing** to be in log mode.
 - It will not shut the interface if the **PD** draws too much power.
 - It will restart the interface and send a **Syslog** message.

PoE - Table Summary

Name	Standard	Watts	Powered Wire Pairs
Cisco Inline Power (ILP)	Cisco Proprietary	7	2
PoE (Type 1)	802.3af	15	2
PoE+ (Type 2)	802.3at	30	2
UPoE (Type 3)	802.3bt	60	4
UPoE+ (Type 4)	802.3bt	100	4

Quality of Service (QoS)

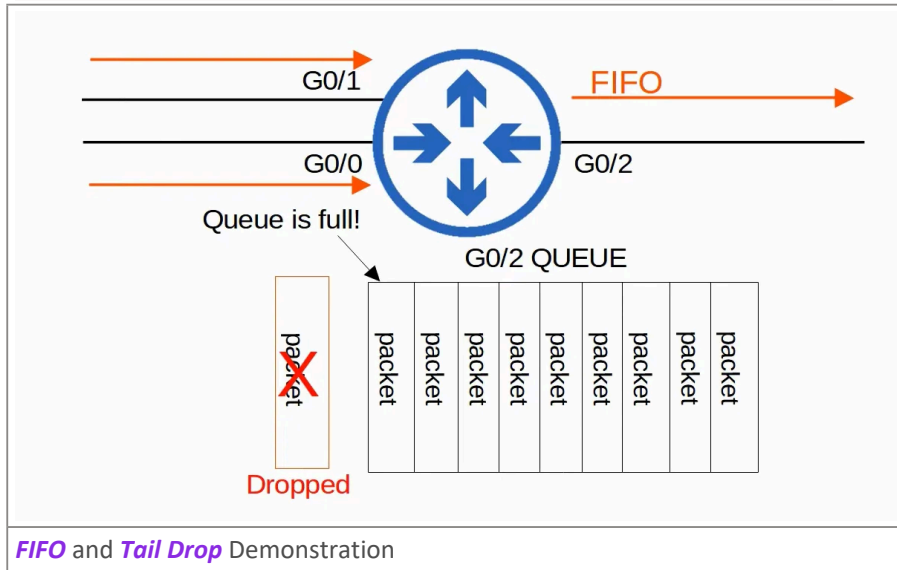
- Modern networks are typically **converged** networks, in which **IP Phones**, video traffic, regular data traffic and etc all share the same IP network.
- This provides more advanced features, however, the different kinds of traffic now have to compete for **bandwidth** on the network. This is where **QoS** comes in.
- **Quality of Service (QoS)** is a set of tools used by network devices to apply different treatments to different packets.
- **QoS** is used to manage the following characteristics of the network traffic:
 1. **Bandwidth**
 - The overall capacity of the link.
 - Measured in **bits per second (Kbps, Mbps, Gbps, etc.)**
 - **QoS** allows you reserve a certain amount of a link's **bandwidth** for specific kinds of traffic.
 2. **Delay**
 - **One-way delay** = The amount of time it takes traffic to go from source to destination.
 - **Two-way delay** = The amount of time it takes traffic to go from source to destination and return back.
 3. **Jitter**
 - The variation in **one-way delay** between packets sent by the same application.
 - **IP Phones** have a '**jitter buffer**' to provide a fixed delay to audio packets.
 4. **Loss**
 - The percent of packets sent that do not reach their destination.
 - Can be caused by faulty cables or when a device's packet **queues** get full and the device starts discarding packets.
- The following standards are recommended for an optimal interactive audio (phone call) quality:
 1. **One-way delay: 150 ms** or less
 2. **Jitter: 30 ms** or less
 3. **Loss: 1%** or less

QoS - Queuing

Queuing is a process that happens when a network device receives messages faster than it can forward them out of the appropriate interface. These messages get placed into a **queue**.

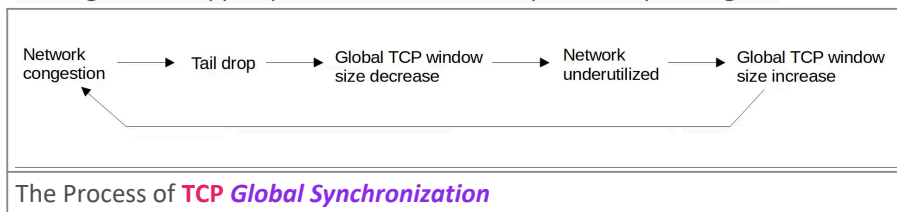
- By default, **queued** messages will be sent out in a **First In First Out (FIFO)** manner. Which means that messages will be sent out in the order they were received in.

- If the **queue** is full, new messages will be dropped. This is called **Tail Drop**.



Tail Drop Explanation

- Tail drop** is harmful because it can lead to **TCP global synchronization**.
Quick review: TCP **sliding window**
 - Hosts using **TCP** use the '**sliding window**' in order to increase/decrease the rate at which they send traffic as needed.
 - When a packet is **dropped**, it will be re-transmitted.
 - When a **drop** occurs, the sender will reduce the **sliding window** size (the rate at which it sends traffic). It will then gradually increase the rate again.
- When the queue fills up and **tail drop** occurs, all the **TCP** hosts sending traffic will slow down the rate at which they send traffic at the same time.
- They will all then gradually increase the rate at which they send traffic, which rapidly leads to more congestion, causing more dropped packets, and the entire process repeats again.



Tail Drop Solution

Some of the solutions that were developed to solve TCP **global synchronization** are '**Random Early Detection (RED)**' and '**Weighted Random Early Detection (WRED)**'

- The way that **RED** works is that when the amount of traffic in the queue reaches a certain threshold, the device will start randomly dropping packets from select **TCP flows**.
- The **TCP flows** that dropped packets will reduce the rate at which traffic is sent, but this will also avoid the **TCP Global Synchronization** problem, in which all **TCP flows** reduce then increase the rate of transmission at the same time in waves.
- In standard **RED**, all kinds of traffic are treated the same (equally).
- An improved version, **Weighted Random Early Detection (WRED)**, allows you to control which packets are dropped depending on the **traffic class (QoS priority)**.